UNIVERSITY OF UYO

DEPARTMENT OF METHEMATICS

MTH313 TEST Advanced Calculus 1 2023/2024 SESSION

Instruction: Answer all the questions Time: 40 minutes

1. (a) If $Z = x^y + y^x$, evaluate $\frac{\partial^2 z}{\partial x \partial y}$. (b) Discuss the existence of the limit of the function $f(x,y) = \frac{xy}{\sqrt{x^2+y^2}}$

2. (a) Consider the periodic function defined by $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$

What is the value of a_n in the Fourier Series expansion of $f(x)$. Hence find the values of a_1, a_2 and a_3.

(b) Find the Laplace transform of $f(t)$ defined by $f(t) = \begin{cases} \frac{t}{k}, & 0 < t < k \\ 1, & t > k \end{cases}$

(b) What set of points in the x-y complex plane satisfies

$$\left| \frac{z}{z-1} \right| = 2$$

CamScanner